UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence — Fall 2025
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Lecture 13

Content.

1. Maximal Inequalities Continued
2. Martingale Concentration Inequalities

1 Maximal Inequalities Continued

Recall the definition of ϵ-net from last lecture.

Definition 1. *(ϵ-Net)* Fix $K \subset \mathbb{R}^d$ and $\epsilon > 0$. A set $N \subset K$ is called an ϵ-net if

$$\forall x \in K,\ \exists x' \in N \text{ such that } \|x - x'\|_2 \leq \epsilon.$$

Lemma 1. For $B_2 := \{x \in \mathbb{R}^d : \sum x_i^2 \leq 1\}$, there exists an ϵ-net N of cardinality at most $\left(1 + \frac{2}{\epsilon}\right)^d$. If $\epsilon \leq 1$, then the bound can be weakened to $(\frac{3}{\epsilon})^d$.

Proof. (Greedy Algorithm) Start with any $x_1 \in B_2$. At step t, we first have $\{x_1, \ldots, x_{t-1}\}$, then we pick any $x_t \notin \{x_1, \ldots, x_{t-1}\}$ such that $\|x_t - x_i\|_2 > \epsilon$ for any $i = 1, ..., t-1$. Let N be the resulting set. We claim that N is an ϵ-net. If not, then there exists $x \in B_2$ such that $\|x - x'\|_2 > \epsilon$ for all $x' \in N$. Then x could have been added to N, a contradiction.

Let $B_x := \{y : \|x - y\| \leq \frac{\epsilon}{2}\}$, $x \in N$. Then we claim that

$$B_x \cap B_y = \emptyset \quad \text{for any } x, y \in N.$$

If false, then there exists $x, y \in N$, and $z \in B_x \cap B_y$. But we have

$$\|z - x\| \leq \frac{\epsilon}{2}, \quad \|z - y\| \leq \frac{\epsilon}{2},$$

which implies

$$\|x - y\| \leq \|z - x\| + \|z - y\| \leq \epsilon,$$

a contradiction against our selection methods of points in N. Also, but a similar triangular inequality argument, we get $\bigcup_{x \in V} B_x \subseteq \{y : \|y\| \leq 1 + \frac{\epsilon}{2}\}$.

Now compute volumes: using the fact $B_x, x \in N$ are disjoint

$$\text{Vol}\left(\bigcup_{x \in N} B_x\right) = |N| \cdot \text{Vol}(B_x) = |N| \cdot \left(\frac{\epsilon}{2}\right)^d \cdot \text{Vol}(B_2),$$

and

$$\text{Vol}\left(\bigcup_{x \in N} B_x\right) \leq \text{Vol}\left(\{y : \|y\| \leq 1 + \frac{\epsilon}{2}\}\right) = \left(1 + \frac{\epsilon}{2}\right)^d \cdot \text{Vol}(B_2).$$

Thus,

$$|N| \cdot \left(\frac{\epsilon}{2}\right)^d \leq \left(1 + \frac{\epsilon}{2}\right)^d \quad \Rightarrow \quad |N| \leq \left(1 + \frac{2}{\epsilon}\right)^d.$$

□

Theorem 1. Let $X \in \mathbb{R}^d$ have i.i.d. sub-Gaussian entries with parameter σ^2. Then

(1)

$$\mathbb{E}\left[\max_{\theta \in B_2} \theta^T X\right] = \mathbb{E}\left[\max_{\theta \in B_2} |\theta^T X|\right] \leq 4\sigma\sqrt{d}.$$

(2)

$$\max_{\theta \in B_2} \theta^T X = \max_{\theta \in B_2} |\theta^T X| \leq 4\sigma\sqrt{d} + 2\sigma\sqrt{\log \frac{1}{\delta}}$$

with probability at least $1 - \delta$.

Proof. (1) Fix M, a $\frac{1}{2}$-net for B_2 with $|M| \leq 6^d$ (we get this from the weakened bound in the above lemma). For any $\theta \in B_2$, there exists $z \in M$ such that $x = \theta - z$ satisfies $\|x\| \leq \frac{1}{2}$. Then

$$\theta^T X = x^T X + z^T X \leq \max_{y \in \frac{1}{2} B_2} y^T X + \max_{z \in M} z^T X,$$

which implies

$$\max_{\theta \in B_2} \theta^T X \leq \max_{y \in \frac{1}{2} B_2} y^T X + \max_{z \in M} z^T X.$$

Note that

$$\max_{y \in \frac{1}{2}B_2} y^T X = \frac{1}{2} \max_{y \in B_2} y^T X.$$

So,

$$\max_{\theta \in B_2} \theta^T X \leq 2 \max_{z \in M} z^T X.$$

Taking expectation:

$$\mathbb{E}\left[\max_{\theta \in B_2} \theta^T X\right] \leq 2\mathbb{E}\left[\max_{z \in M} z^T X\right].$$

From last lecture,

$$2\mathbb{E}[\max_{z \in M} z^T X] \leq 2\sigma\sqrt{2\log|M|} \leq 2\sigma\sqrt{2\log 6 \cdot d} \leq 4\sigma\sqrt{d},$$

which is the desired result.

(2) We have:

$$\left\{\max_{\theta \in B_2} \theta^T X > t\right\} \subset \left\{\max_{z \in M} z^T X > \frac{t}{2}\right\}.$$

Then,

$$\mathbb{P}\left(\max_{\theta \in B_2} \theta^T X > t\right) \leq \mathbb{P}\left(\max_{z \in M} z^T X > \frac{t}{2}\right) \leq |M| \cdot \mathbb{P}\left(z^T X > \frac{t}{2}\right),$$

where the last inequality is by the union bound. Using sub-Gaussian tail bound:

$$\mathbb{P}\left(z^T X > \frac{t}{2}\right) \leq e^{-\frac{t^2}{8\sigma^2}}.$$

So,

$$\mathbb{P}\left(\max_{\theta \in B_2} \theta^T X > t\right) \leq 6^d e^{-\frac{t^2}{8\sigma^2}}.$$

Set this equal to δ, then simplify it to get the result. □

2 Martingale Concentration Inequalities

Definition 2 (Martingale)**.** Let $Z_1, \ldots, Z_n, \ldots$ be random variables. A stochastic process $(X_n)_{n \geq 0}$ is called a martingale if

(a) X_n is measurable w.r.t. $\sigma(Z_1, \ldots, Z_n)$,

(b) $\mathbb{E}[|X_n|] < \infty$, and

(c) $\mathbb{E}[X_{n+1} \mid Z_1, \ldots, Z_n] = X_n$.

Definition 3 (Doob Martingale)**.** Let $f : \mathbb{R}^N \to \mathbb{R}$. Define

$$X_n = \mathbb{E}[f(Z_1, \ldots, Z_N) \mid Z_1, \ldots, Z_n], \quad n = 0, 1, \ldots, N.$$

Then $(X_n)_{n=0,\ldots,N}$ is called a Doob martingale (assume $\mathbb{E}|f| < \infty$).

Proof. (Doob martingale is a martingale)

$$\begin{aligned}
\mathbb{E}[X_{n+1} \mid Z_1, \ldots, Z_n] &= \mathbb{E}\left[\mathbb{E}[f \mid Z_1, \ldots, Z_{n+1}] \mid Z_1, \ldots, Z_n\right] \\
&= \mathbb{E}[f \mid Z_1, \ldots, Z_n] \\
&= X_n.
\end{aligned}$$

□

Let $(X_n)_{n\geq 1}$ be a martingale with respect to a filtration $\mathcal{F}_n$, and $X_0 = 0$ so that $\mathbb{E}X_0 = \mathbb{E}X_1 = ... = 0$ (since $\mathbb{E}X_n = \mathbb{E}[X_{n+1}|\mathcal{F}_n] = \mathbb{E}X_{n+1}$). Our goal is to find bounds of the form

$$\mathbb{P}[|X_n| \geq t] = \mathbb{P}[|X_n - \mathbb{E}X_n| \geq t] \leq \exp(-\Theta(n)).$$

Theorem 2. Let $(X_n)_{n\geq 1}$ be a martingale w.r.t. filtration $\mathcal{F}_n$, with $X_0 = 0$. Suppose $\exists d_1, \ldots, d_n$ such that

$$|X_i - X_{i-1}| \leq d_i \quad \text{a.s.}, \quad i = 1, \ldots, n.$$

Then,

$$\mathbb{P}(|X_n| \geq t) \leq 2\exp\left(-\frac{t^2}{2\sum_{i=1}^n d_i^2}\right).$$

Remark. It applies to potentially dependent processes. The only requirement is that increments are almost surely bounded.

Remark. The idea of the proof is still similar to that for the Chernoff bound. We start with

$$\mathbb{P}(X_n \geq t) = \mathbb{P}(e^{\lambda X_n} \geq e^{\lambda t}) \leq e^{-\lambda t}\mathbb{E}[e^{\lambda X_n}].$$

Then control $\mathbb{E}[e^{\lambda X_n}]$ by writing $X_n = \sum_{i=1}^n (X_i - X_{i-1})$.

Lemma 2. Fix $\lambda \in \mathbb{R}$. Then for any $x \in [-d_i, d_i]$,

$$e^{\lambda x} \leq \frac{e^{\lambda d_i} + e^{-\lambda d_i}}{2} + \frac{e^{\lambda d_i} - e^{-\lambda d_i}}{2d_i} x.$$

Proof. Fix $x \in [-d_i, d_i]$. Write $x = \theta(-d_i) + (1-\theta)d_i$, $\theta \in [0,1]$, then

$$\theta = \frac{d_i - x}{2d_i} = \frac{1}{2}\left(1 - \frac{x}{d_i}\right).$$

Since $f(x) = e^{\lambda x}$ is convex, we have

$$f(x) \le \theta f(-d_i) + (1-\theta) f(d_i) = \frac{1}{2}\left(1 - \frac{x}{d_i}\right) e^{-\lambda d_i} + \frac{1}{2}\left(1 + \frac{x}{d_i}\right) e^{\lambda d_i}.$$

Simplify to get the result. □